

COS80023 – Big Data

Distinction/High Distinction Report

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Arun Ragavendhar Arunachalam Palaniyappan – 104837257

Vikas Kadruvan – 104981442

Nikolaos Anestakis – 104014513

Tutor: Dr. Afzaal Hassan

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School of Science, Computing and Engineering Technologies

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Contribution statements

Student name	Student ID	Contributions
Arun Ragavendhar Arunachalam Palaniyappan	104837257	I was the team leader. I designed and drafted the project idea and planned, discussed the scope with the team members, and allocated tasks. I authored all the subheadings under Section 2.1. Why are accidents in Rural Victoria deadlier than other regions in the last 14 years, and what are the reasons behind this? Do Motorcyclists and Heavy Vehicles Really Cause Most Crashes, as most people perceive? Also, I set the report outline, prepared initial drafts, and after everyone added their parts I edited, formatted, and finalised the document and made it ready for submission.
Vikas Kadruvan	104981442	I authored the following section and all the subheading under the section 2.2. How have crash mechanisms, environmental conditions, and road-user groups jointly influenced crash patterns and safety outcomes across Victorian regions from 2012–2025, Documentation, Summary, Introduction, Conclusion. I proofread the report before submission.
Nikolaos Anestakis	104014513	I authored Section 2.3 and all its subheadings. Why do Frankston, Kingston, and Colac Otway still record severe crashes despite state-wide safety gains, and which spatial, behavioural, age-related, or environmental factors explain this persistence? I addressed review comments on Section 2.3 and kept the writing consistent with the report style. I helped with formatting the report.

Executive Summary

This study looked at the combined effects of geographic, environmental, mechanical, and human factors on crash persistence and severity throughout Victoria between 2012 and 2025. The study challenges preconceived notions about the types and conditions of vehicles that cause most crashes and explains why rural areas continue to be deadlier, making it useful for policymakers and the public. ACCIDENT_NO was used to join, clean, standardise, and aggregate multi-table crash data from VicRoads (accident, node, person, vehicle, road surface, and atmospheric files) into five time periods. R was used to create analytical visualisations (ggplot2, dplyr, leaflet) to investigate demographic, temporal, and spatial trends.

The findings indicate that high-speed (≥ 90 km/h) mid-block and T-intersection crashes, poor surface conditions, and heavy-vehicle involvement act as severity multipliers rather than primary causes in rural Victoria's fatality rates. While regional zones exhibit severity-based risks that are increased by road geometry and maintenance quality, urban corridors face frequency-based risks. The three persistent hotspots, Frankston, Kingston, and Colac Otway, reflect various environmental and behavioural problems like traffic, lighting, and fatigue.

The results offer compelling evidence for focused, context-specific safety interventions rather than broad, uniform measures, even though data limitations (missing exposure data, underreporting, and post-COVID anomalies) limit absolute generalisation.

1. Introduction

Despite statewide improvements, road safety is still one of Victoria's most urgent public concerns, with rural areas still experiencing disproportionately severe crashes. This study examines how speed, road geometry, surface conditions, and user groups interact to influence crash severity to uncover the underlying structural and environmental causes of these results. The study uses multi-year Victorian crash datasets and sophisticated visual analytics in R to find hidden behavioural and spatial patterns that explain why some places and circumstances continue to be high-risk.

2. Key Insights and Analysis

2.1 Why are accidents in Rural Victoria deadlier than other regions in the last 14 years, and what are the reasons behind this? Do Motorcyclists and Heavy Vehicles Really Cause Most Crashes, as most people perceive?

The comparison of fatality rates across different geographic regions in Victoria, shows that rural Victoria consistently records higher rates than other regions. This study looks to find out which

combinations of road geometry, speed, weather, road surface, and lighting explain the higher fatality risk in rural Victoria over the last 14 years.

It also tests whether the popular public perception, that motorcycles or heavy vehicles are the main reasons for most road accidents, are true or not, once exposure and severity data are considered. The analysis compares patterns across regions and time and looks at interactions between factors rather than single causes, to get deeper insights, which can then be used to suggest mitigation actions.

deg_urban_name	count
MELB_URBAN	114603
RURAL_VICTORIA	39791
LARGE_PROVINCIAL_CITIES	10479
SMALL_CITIES	9401
TOWNS	5618
MELBOURNE_CBD	2219
SMALL_TOWNS	1814
NA	903

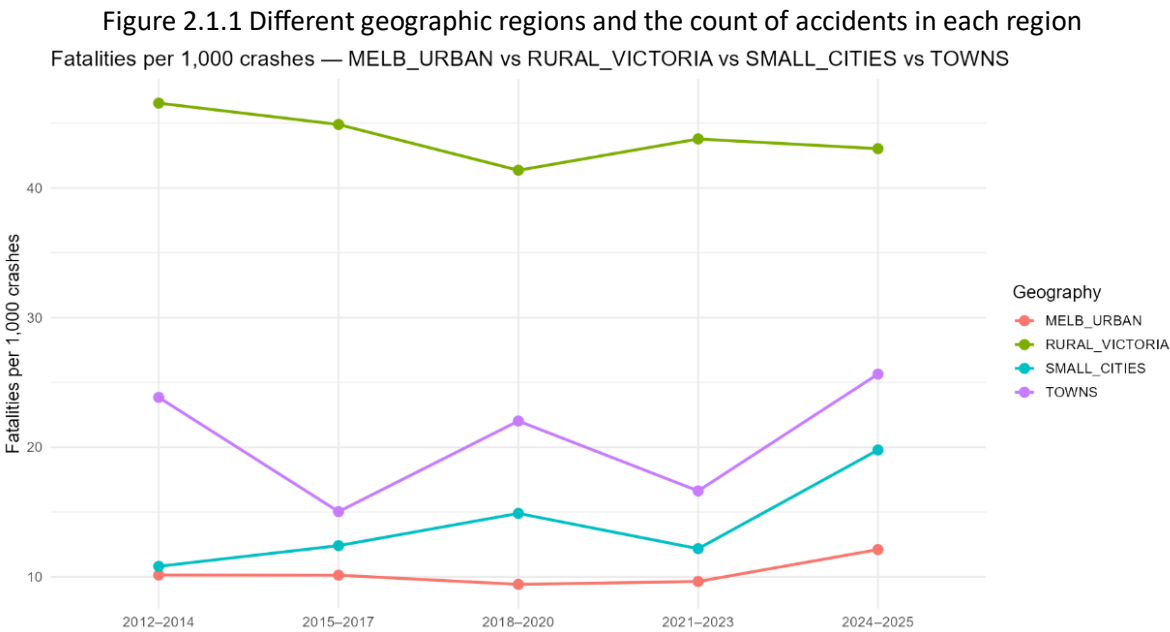


Figure 2.1.2 Fatality rates in different geographic regions in different time periods

2.1.1 Methodology

The study joins, filters and aggregates the given multi-year Victorian crash data from accident.csv, node.csv, road_surface_cond.csv, atmospheric_cond.csv, person.csv, lighting.csv and vehicle.csv, joining each on the key ACCIDENT_NO. Dates were parsed, years derived, and crashes grouped into five periods (2012–14, 2015–17, 2018–20, 2021–23, 2024–25). Rows without a period or region were removed. Regions with significant number of accidents were filtered and the regions MELB_URBAN, RURAL_VICTORIA, SMALL_CITIES, and TOWNS, were chosen, keeping the crash as the unit of analysis while preserving links to people and vehicles. Simple categories were

prepared for posted speed, road geometry, lighting, weather, and road surface. Motorcycle and heavy-vehicle involvement was flagged from vehicle.csv using Vehicle type descriptions and mass or axle values, then rolled up to the crash level. For each period × region × factor, crashes, fatalities, fatalities per 1,000 crashes, and share of crashes were calculated. All steps were completed in R (dplyr, ggplot2), and the figures and matching CSVs were saved.

2.1.1.1. Data Preparation

- Period definition: The crash year was mapped to five fixed intervals: 2012–14, 2015–17, 2018–20, 2021–23, 2024–25.
- Region mapping: node.csv (DEG_URBAN_NAME) was mapped to four regions, MELB_URBAN, RURAL_VICTORIA, SMALL_CITIES, TOWNS.
- Key and cleaning: ACCIDENT_NO was set as the join key; dates were parsed (including Excel serials); rows outside the five periods or without a region were removed.
- Condition joins: Road surface and weather fields were attached by joining the relevant tables to the crash records on ACCIDENT_NO.
- Road-user flags: vehicle.csv was used to flag motorcycle-involved and heavy-vehicle-involved crashes (using vehicle type descriptions and mass/axle values), then rolled up to crash level.
- Standard bins: Comparable buckets were created for
 - Posted speed (≤ 60 , 70–80, ≥ 90 km/h)
 - Road geometry (Not at intersection, Cross intersection, T intersection)
 - Lighting (Daylight, Dusk/Dawn, Dark-Lighted, Dark-Unlighted)
 - Road surface (Dry, Wet, Loose/Gravel, Snow/Ice, Oil/Mud/Debris)
 - Weather (Fine/Clear, Raining, Fog/Smoke, Snow/Hail, Windy, Dust)
- Derived metrics: For each period × region × factor, the crashes, fatalities, fatalities per 1,000 crashes were computed, and share of crashes.

The screenshot displays the R Studio interface. The left pane shows the R script editor with the following code:

```

32 num <- suppressWarnings(as.numeric(x_chr)) # Excel serials
33 need_serial <- is.na(dt) & !is.na(num)
34 if(any(need_serial)) dt[need_serial] <- as.Date(num[need_serial], origin = "1899-12-30")
35 as.Date(dt)
36 }
37
38 label_period <- function(y){
39   dplyr::case_when(
40     y %in% 2012:2014 ~ "2012-2014",
41     y %in% 2015:2017 ~ "2015-2017",
42     y %in% 2018:2020 ~ "2018-2020",
43     y %in% 2021:2023 ~ "2021-2023",
44     y %in% 2024:2025 ~ "2024-2025",
45     TRUE ~ NA_character_
46   )
47 }
48 years_in_period <- c("2012-2014"=3,"2015-2017"=3,"2018-2020"=3,"2021-2023"=3,"2024-2025"=2)
49 period_order <- names(years_in_period)
50
51 # 2) Load data (CSV files in /data) -----
52 acc_path <- "data/accident.csv"
53 node_path <- "data/node.csv"
54
55 stopifnot(file.exists(acc_path), file.exists(node_path))
56
57 acc <- readr::read_csv(acc_path, show_col_types = FALSE) %>% janitor::clean_names()
58 node <- readr::read_csv(node_path, show_col_types = FALSE) %>% janitor::clean_names()
59
60 # Basic column checks (fail fast with a clear message)
61
11674 4) Baseline severity per 1,000 by period - Geo4

```

The right pane shows the 'Global Environment' with a list of data objects:

Object	Size	Variables
acc	182039 obs.	23 variables
acc_comp	169413 obs.	40 variables
acc_drv	169413 obs.	35 variables
acc_harm	183089 obs.	37 variables
acc_nodes	183089 obs.	38 variables
acc4	169413 obs.	26 variables
acc4_geom	169413 obs.	28 variables
acc4_hv	169413 obs.	32 variables
acc4_light	169413 obs.	27 variables
acc4_mc	169413 obs.	32 variables
acc4_mc_f	152834 obs.	32 variables
acc4_speed	169413 obs.	28 variables
acc4_surface	170379 obs.	28 variables
acc4_time	0 obs.	33 variables
acc4_weather	169413 obs.	27 variables
baseline_geo4	20 obs.	7 variables
car_sum	20 obs.	8 variables
cmp	45 obs.	7 variables
cmp2	45 obs.	8 variables
comp_flags	182039 obs.	9 variables
deg_counts_raw	1 obs.	1 variable
df	105 obs.	3 variables
df_g	4523 obs.	19 variables
dodge	Environment	
drv_exp	0 obs.	7 variables
drv_flags	0 obs.	4 variables

The bottom pane shows the console output:

```

> gggsave("output/insight_baseline_geo4.png", p_base4, width = 12, height = 6, dpi = 150)
> message("Baseline complete. Files written to /output")
Baseline complete. Files written to /output

```

Figure 2.1.1.1.1 Code getting executed in R studio

2.1.1.2. Visualisation

The visualisation uses RStudio (ggplot2, dplyr, scales). Baseline line charts show fatalities per 1,000 crashes by period and region. Grouped bar charts compare posted speed, road geometry, lighting, weather, and road surface across regions and periods. Scatterplots place share of crashes (%) on the x-axis and fatalities per 1,000 on the y-axis to separate how common something is from how deadly it is. Motorcycle and heavy-vehicle views repeat the same setup.

2.1.2. Results

Speed

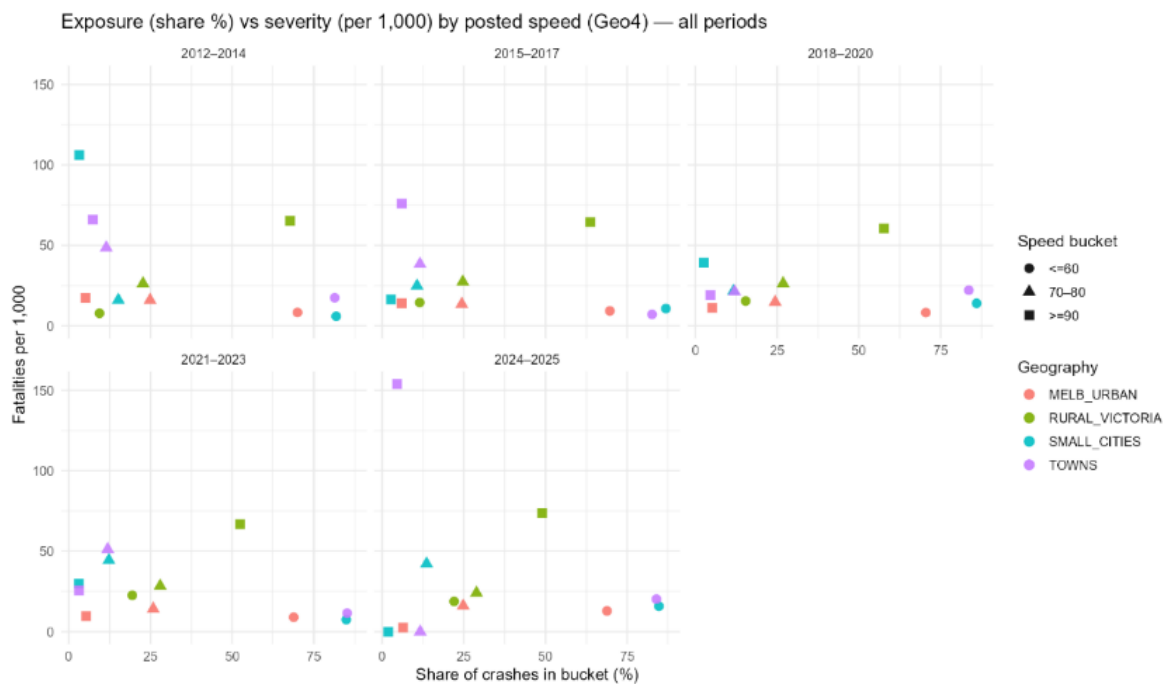


Figure 2.1.2.1 Exposure (share %) vs severity (per 1000) by posted speed

The above scatterplot shows a consistent pattern across all periods. Roads signed ≥ 90 km/h in rural Victoria carry the highest fatality rates (45-75 fatalities per 1000) and account for a major share of the crashes (more than 50%). This is not case for other regions, as the fatality rate on roads signed ≥ 90 km/h is very less (5-20 fatalities per 1000) with a very low share of the crashes (less than 10%).

This means that speeding above 90 km/h on rural Victoria roads have twice or at times thrice the probability of a crash being fatal, when compared to other regions.

Also, small cities and towns show similar spikes at ≥ 90 km/h in some periods but with smaller exposure. These results set up the next checks on road geometry, lighting, surface, and weather to see which conditions combine with high speed, to push risk even higher.

Road Geometry

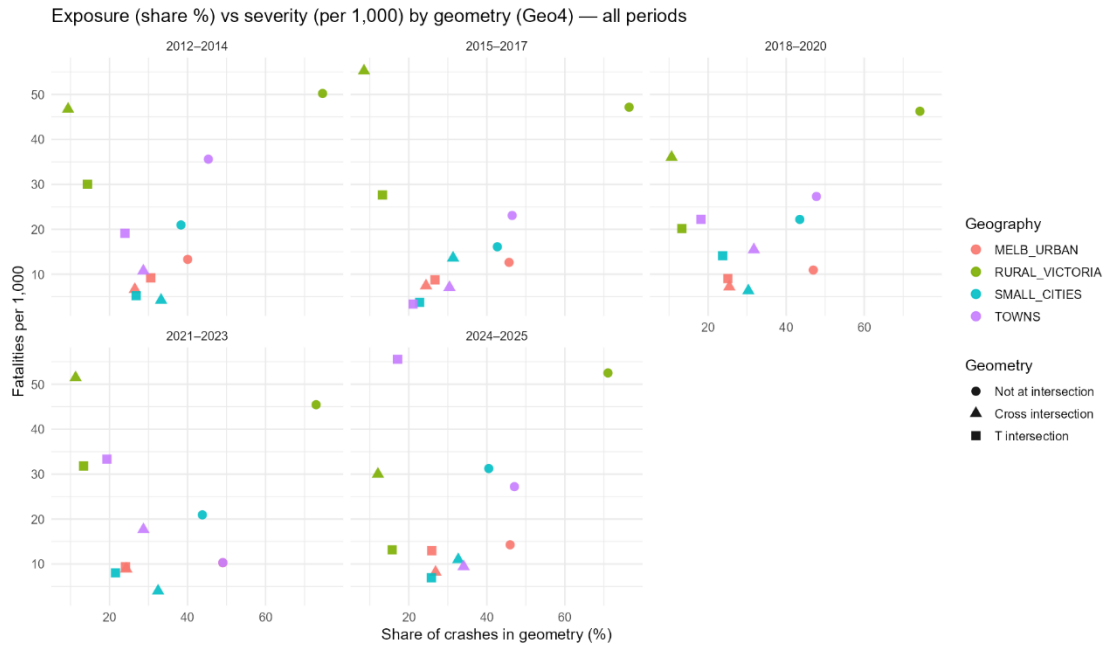


Figure 2.1.2.2 Exposure (share %) vs severity (per 1000) by road geometry

The road geometry results show that in rural Victoria, mid-blocks (open-road links between junctions, such as straight or gentle curves, crests or dips, overtaking sections, bridge approaches) carry fatality rates around 45-60 per 1,000 and a high share of crashes. T-intersections are also high and, in some periods, exceed mid-blocks. Other regions do not show this combination of high rate and high share. Combined with the speed view, the specific high-risk case is ≥ 90 km/h on rural Victoria's mid-blocks and at T-intersections, a pattern not seen in other regions, like Melbourne's urban area, small cities, or towns. This sets up the next checks on lighting, surface, and weather to see which conditions add to this geometry-speed mix.

Road surface Conditions

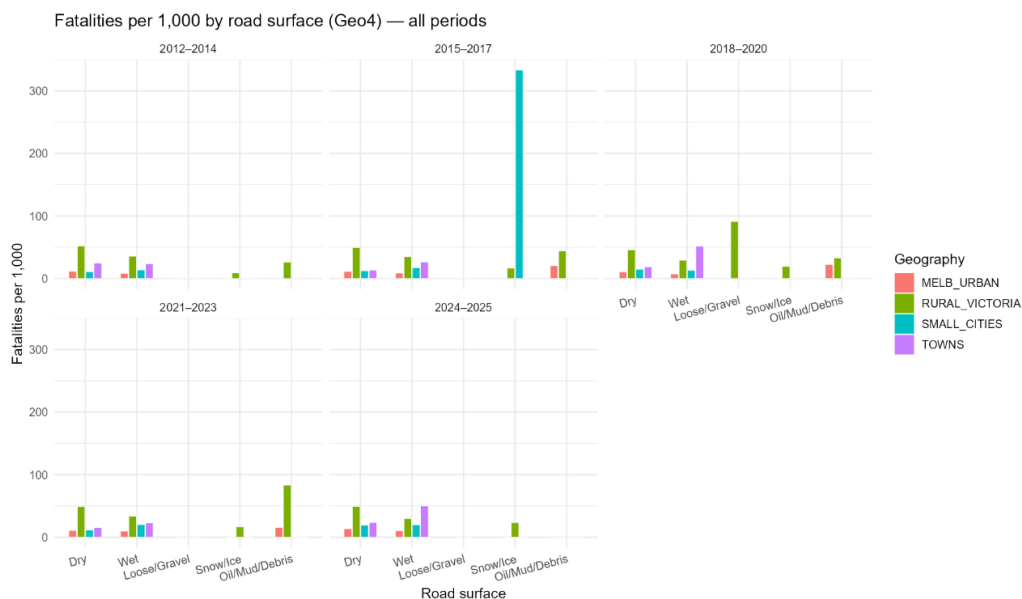


Figure 2.1.2.3 Fatalities (per 1000) by road surface conditions

The road-surface results show a one-off spike in small cities under snow/ice in 2015–2017. After that, this condition does not feature with notable fatalities. In rural Victoria, which is the focus, two surfaces are consistently the highest on cumulative fatality rate: loose/gravel and oil/mud/debris; dry and wet surfaces also sit high. Taken together, fatality rates are higher across most surface conditions in rural Victoria, in comparison to other regions, pointing to a broader issue with road quality, specific to rural Victoria. When this poorer surface standard combines with high speeds and slopes/curves on mid-blocks (open-road links between junctions), the risk compounds, helping explain the region’s high fatality rate. The next step checks weather conditions and lighting conditions to see how they interact with these surfaces.

Weather Conditions

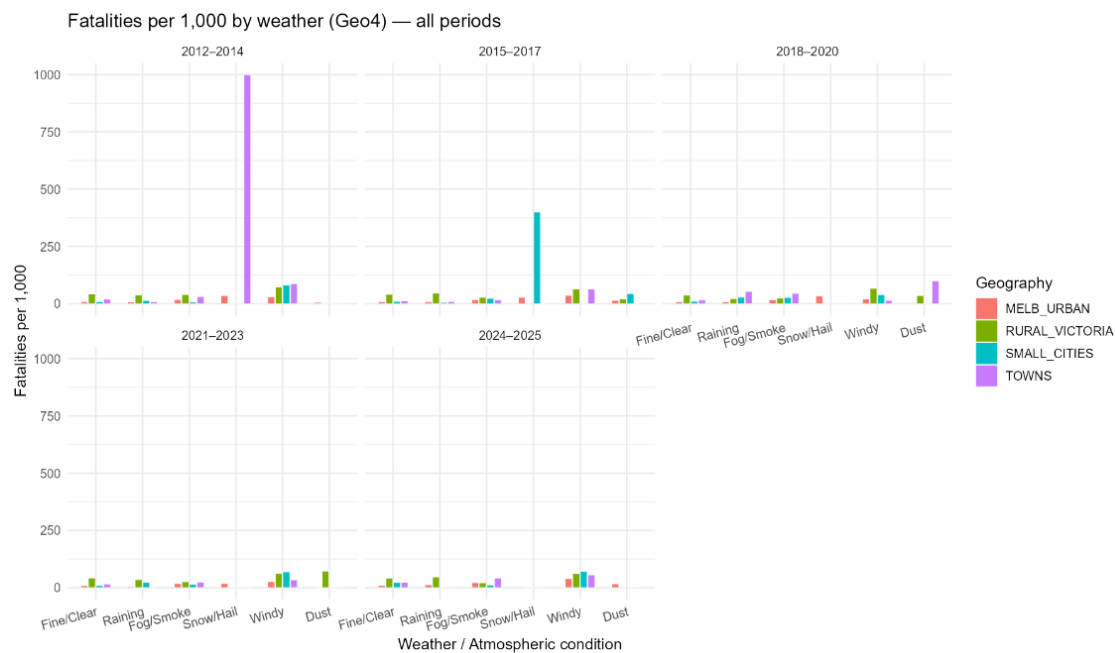


Figure 2.1.2.4 Fatalities (per 1000) by weather conditions

The weather view shows little sustained effect on fatal risk except in rare snow/hail cases. Two one-off spikes appear in the plots, towns in 2012–2014 at roughly 900 fatalities per 1,000 and small cities in 2015–2017 around 250 - 400 per 1,000, but neither reappears in later periods. Across the other bands, which are, fine/clear, rain, fog/smoke, windy, dusty, rates stay relatively low and steady, including in rural Victoria. This indicates the rural gap is not driven by everyday weather, rather, aligns more with high speeds, mid-block/T-intersection geometry, and surface quality, with snow only acting as an occasional amplifier where it occurs.

Heavy Vehicles in crashes

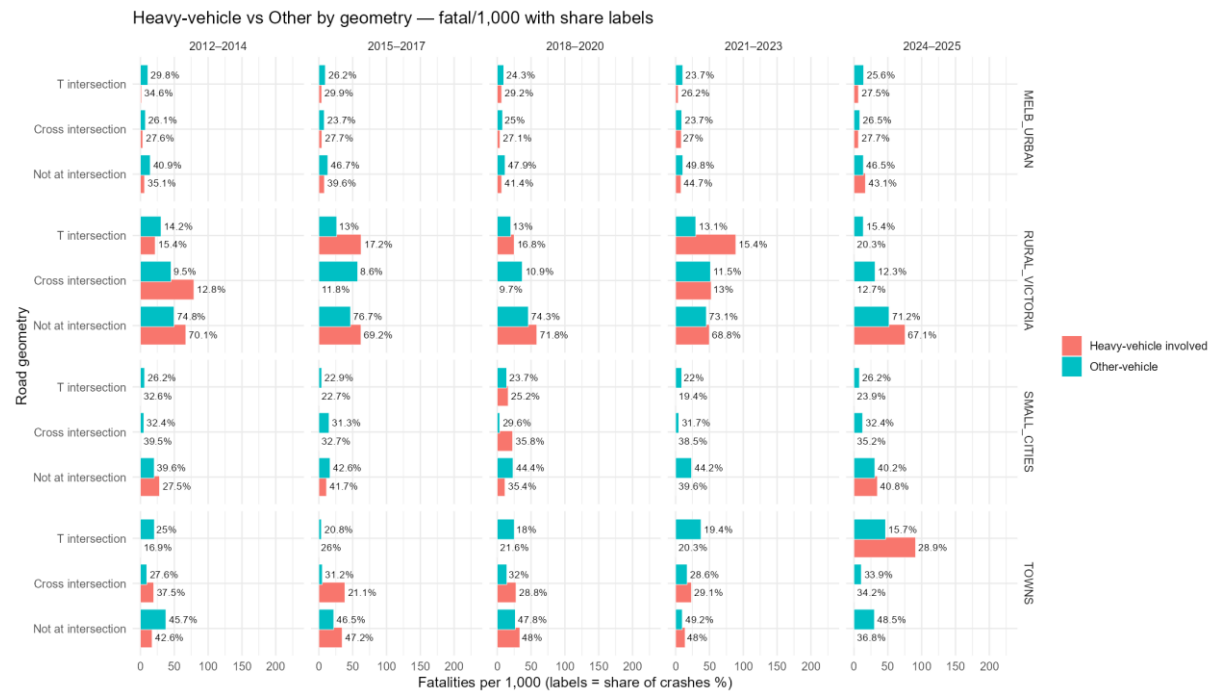


Figure 2.1.2.5 Heavy vehicles in crashes (Fatality rate, involvement vs other vehicles, road geometry)

The widespread public perception is that heavy vehicles cause most crashes. But the data shows this is untrue and adds an important nuance to check further. Across regions and periods, other vehicles in fact account for more crashes, but when heavy vehicles are involved, the fatality rate is higher. In rural Victoria these crashes cluster on mid-blocks (open-road links between junctions) and next at T-intersections. In these settings, their fatality rates typically exceed those of other vehicles, and mid-blocks carry roughly 70-75% of heavy-vehicle crashes across the timeline, while metro Melbourne, small cities, and towns sit lower on both rate and share. So, even though heavy vehicles act as severity multipliers, they are not the main source of crash counts.

Motorcycles in Crashes

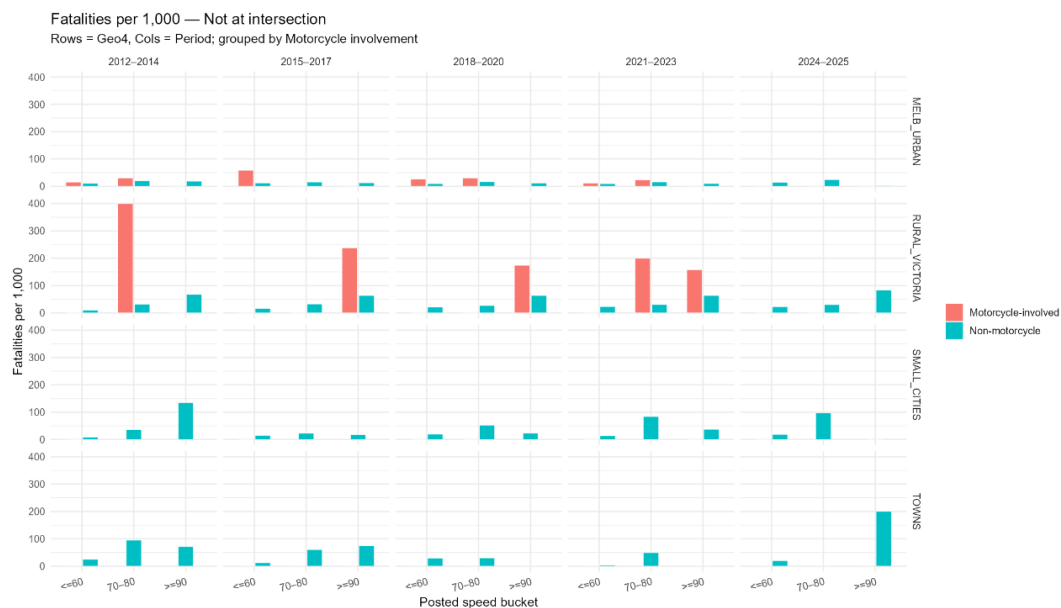


Figure 2.1.2.6 Fatality rate of motorcycles in every speed band in every region (T intersections)

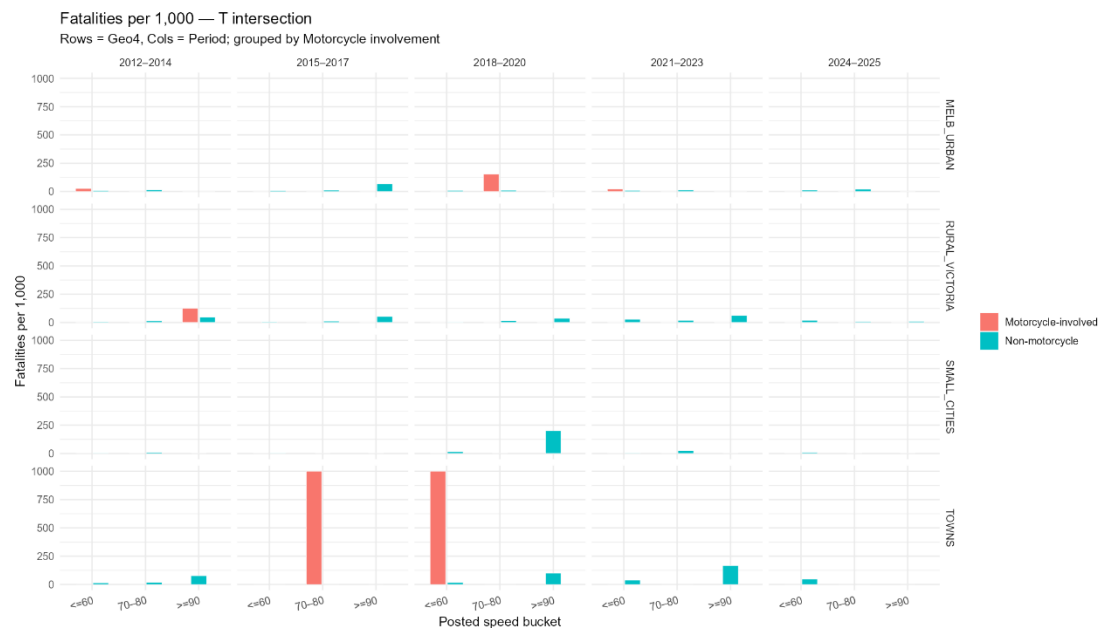


Figure 2.1.2.7 Fatality rate of motorcycles in every speed band in every region(midblock)

The motorcycle view challenges the common claim that motorcyclists cause most crashes or mainly crash and cause nuisance at extreme speeds. Across regions and periods, other vehicles account for more crashes. When motorcycles are involved, the fatality rate is higher, but the concentration sits in the 70-80 km/h band on mid-blocks and at T-intersections, not in the ≥ 90 km/h band. Several periods show the 70-80 band above the ≥ 90 band for motorcycle fatality rate, while the overall share of motorcycle crashes remains well below cars and other vehicles. This also counters the belief that “the faster the rider, the greater the danger”. Motorcycle crashes seem to be fewer on open high-speed roads, and the higher fatality is seen where riders mix with traffic at 70-80 km/h, often boxed in by cars and trucks, which drives both crash counts and severity in that band.

Final Inference

Crashes in rural Victoria are deadlier because the worst harm happens on high-speed (≥ 90 km/h) rural mid-blocks and T-intersections. Fatality rates sit around 45-75 deaths per 1,000 crashes and this speed bucket makes up over 50% of crashes there. In other regions the same speed sits near 5–20 deaths per 1,000 crashes with under 15% share. Within rural layouts, mid-blocks sections average 45-60 deaths per 1,000 crashes and T-intersections are also high. The Road surface makes this worse, as fatality rates on loose/gravel and oil/mud/debris are the highest, and even fatality rates in dry and wet are higher than elsewhere. Everyday weather is not the driving reason, and rare snow/hail spikes appear outside rural Victoria, in other regions and do not last. Also, the idea that one group “causes most crashes” is not supported. The other vehicles make up more crashes overall, while heavy vehicles raise severity where they appear and cluster on rural mid-blocks (about 70-75% of heavy-vehicle crashes), acting as severity multipliers rather than the main source of crashes. Motorcycles are a smaller share, with higher fatality mostly at 70–80 km/h in mixed traffic, not at ≥ 90 km/h. This challenges the idea that faster riding automatically means greater danger. In all, rural Victoria’s higher fatality rate is mainly due to

very high speeds on open-road sections between junctions (mid-blocks) and at T-intersections, together with poorer road surfaces. Everyday weather is not the cause, and no single road-user group is to blame.

2.1.3. Mitigation Strategies to fix these issues

Main issue	Planned actions (plain and actionable)
High speed on rural mid-blocks (≥90 km/h)	Authorities can lower speed limits where harm is high (to 70 or 60 km/h), deploy average-speed cameras, widen and seal shoulders, and add rumble strips.
Fast approaches to rural T-intersections	Authorities can add turning lanes, slow traffic on approach (roundabout or raised platform where feasible), clear sight lines, and improve signs and markings.
Poor or loose/wet surfaces	Road maintenance teams can relay or seal loose gravel, renew worn surfaces, clear oil/mud/debris quickly, and improve drainage to prevent standing water.
Freight routes (heavy vehicles)	Enforcement can target speed and road safety officers can add overtaking lanes, manage driveway access, widen shoulders, add rumble strips, and space rest areas to reduce fatigue.
70–80 km/h motorcycle corridors (mixed traffic)	Authorities can remove squeeze points, repair edge defects, smooth surfaces, and add barriers on sharp curves where warranted.
Repeat harm sites (blackspots)	Programs can fix the exact problem observed, start with quick low-cost works, escalate to layout changes if needed, and measure before/after.

Table 2.1.3.1 Mitigation strategies

2.2. How have crash mechanisms, environmental conditions, and road-user groups jointly influenced crash patterns and safety outcomes across Victorian regions from 2012–2025

This integrated analysis looks at how road-user demographics, environmental surface conditions, and mechanical crash types have all influenced crash patterns and safety outcomes throughout Victorian regions between 2012 and 2025. The analysis offers a multifactorial view of crash causation by combining several relational datasets, such as accident, road surface, node, and person data. It finds persistent structural risks and temporal anomalies in Victoria's road-safety environment using descriptive and comparative visual analytics.

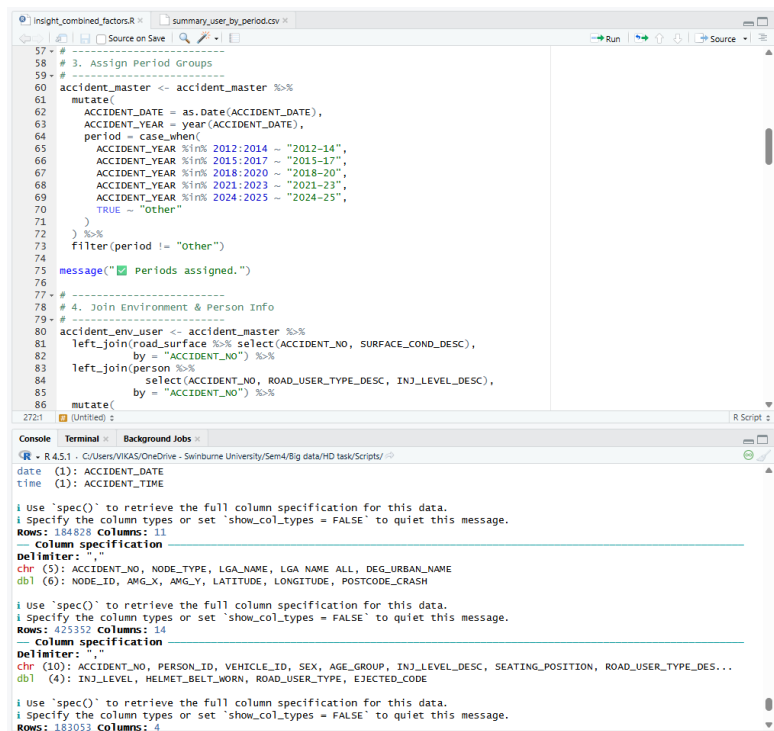
2.2.1. Methodology

The accident.csv, road_surface_cond.csv, node.csv, and person.csv tables—all connected by the primary key ACCIDENT_NO—were first used to extract datasets. Together, these datasets recorded participant demographics, environmental factors, crash characteristics, and location context. To clean up the data, null entries were eliminated, inconsistent column headers were renamed, and derived fields were created for period grouping (2012–14, 2015–17, 2018–20, 2021–23, 2024–25).

Using the dplyr pipeline, the cleaned tables were combined in R. One-to-many and many-to-many connections between crashes and participants were maintained by each join operation. Then, to capture mechanical, environmental, and human-factor patterns, four structured visualisations were created using ggplot2.

2.2.1.1. Data Preparation

- **Period Definition:** For consistent temporal comparison, a year variable was mapped to five fixed intervals (2012–14–2024–25).
- **Crash Type & Region Mapping:** To categorise crash mechanisms according to urbanisation level, ACCIDENT_TYPE_DESC was paired with DEG_URBAN_NAME from node.csv.
- **Environmental Integration:** To attach SURFACE_COND_DESC to every crash record, the road_surface_cond.csv table was joined using ACCIDENT_NO.
- **Road-User Extraction:** The individual. The CSV table was combined with severity indicators from the accident table after being filtered by user category (ROAD_USER_TYPE_DESC).
- **Derived Metrics:** Group_by () and summarise () functions were used to aggregate calculated proportions (share, rate, or fatality percentage) by region and time-period.



```

57 # -----
58 # 3. Assign Period Groups
59 # -----
60 accident_master <- accident_master %>%
61   mutate(
62     ACCIDENT_DATE = as.Date(ACCIDENT_DATE),
63     ACCIDENT_YEAR = year(ACCIDENT_DATE),
64     period = case_when(
65       ACCIDENT_YEAR %in% 2012:2014 ~ "2012-14",
66       ACCIDENT_YEAR %in% 2015:2017 ~ "2015-17",
67       ACCIDENT_YEAR %in% 2018:2020 ~ "2018-20",
68       ACCIDENT_YEAR %in% 2021:2023 ~ "2021-23",
69       ACCIDENT_YEAR %in% 2024:2025 ~ "2024-25",
70       TRUE ~ "Other"
71     )
72   ) %>%
73   filter(period != "Other")
74
75 message("Periods assigned.")
76
77 # -----
78 # 4. Join Environment & Person Info
79 # -----
80 accident_env_user <- accident_master %>%
81   left_join(road_surface %>% select(ACCIDENT_NO, SURFACE_COND_DESC),
82     by = "ACCIDENT_NO") %>%
83   left_join(person %>%
84     select(ACCIDENT_NO, ROAD_USER_TYPE_DESC, INJ_LEVEL_DESC),
85     by = "ACCIDENT_NO") %>%
86   mutate(
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date (1): ACCIDENT_DATE
time (1): ACCIDENT_TIME

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i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
Rows: 184828 Columns: 11
--- column specification ---
Delimiter: ","
chr (5): ACCIDENT_NO, NODE_TYPE, LGA_NAME, LGA_NAME ALL, DEG_URBAN_NAME
dbl (6): NODE_ID, AMG_X, AMG_Y, LATITUDE, LONGITUDE, POSTCODE_CRASH

i Use 'spec()' to retrieve the full column specification for this data.
i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
Rows: 425352 Columns: 14
--- column specification ---
Delimiter: ","
chr (10): ACCIDENT_NO, PERSON_ID, VEHICLE_ID, SEX, AGE_GROUP, INJ_LEVEL_DESC, SEATING_POSITION, ROAD_USER_TYPE_DESC...
dbl (4): INJ_LEVEL, HELMET_BELT_WORN, ROAD_USER_TYPE, EJECTED_CODE

i Use 'spec()' to retrieve the full column specification for this data.
i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
Rows: 183053 Columns: 4

```

Figure 2.2.1.1.1 Codebase for Data preparation

2.2.1.2. Visualisation

After the data was prepared, RStudio's ggplot2, dplyr, and scales packages were used to implement an organised visual analytics procedure. Each visualisation was created to convey a different analytical dimension: human (road-user type), environmental (surface conditions), and mechanical (crash type), all of which together show how these elements interact to affect severity outcomes. Instead of using predictive modelling, the analysis was only descriptive and comparative, concentrating on multi-factor aggregation.

2.2.2. Results

This cohesive analytical framework that demonstrated temporal and cross-factor variations in crash outcomes throughout Victoria was generated by the multi-dimensional visualisation approach. The study discovered structural risk patterns that would not have been apparent through raw frequency analysis alone by normalising each dataset, connecting environmental and behavioural variables, and displaying their interactions through comparative and proportion-based plots. To explain how crash mechanisms, environmental surface conditions, and road-user groups collectively shaped fatality rates and regional safety disparities between 2012 and 2025, the following Results section interprets these visuals collectively.

The combined analysis shows the close relationship between crash mechanisms, environmental factors, and human exposure patterns. The overall frequency of crashes decreased slightly after 2017, but the proportions of fatalities stayed the same, indicating ongoing structural risks. The urban areas of Melbourne and rural Victoria showed clear temporal and regional differences.

Crash Type and Regional Patterns

The distribution of accident mechanisms (such as collisions with vehicles, struck pedestrians, and collisions with fixed objects) across Victoria's urbanisation levels and time periods was examined in this visualisation. Joining accident was the source of the data.csv and node.csv on ACCIDENT_NO, followed by grouping by DEG_URBAN_NAME, ACCIDENT_TYPE_DESC, and five standardised periods (2012–14, 2015–17, 2018–20, 2021–23, 2024–25).

Comparative distributions over time were shown using a faceted column chart (geom_col() with facet_wrap(~period)). To account for uneven crash counts annually, the chart used normalised proportions (share of total crashes). Coord_flip() was used during testing to prevent label clutter, and colors were set to a uniform palette for regional identity.

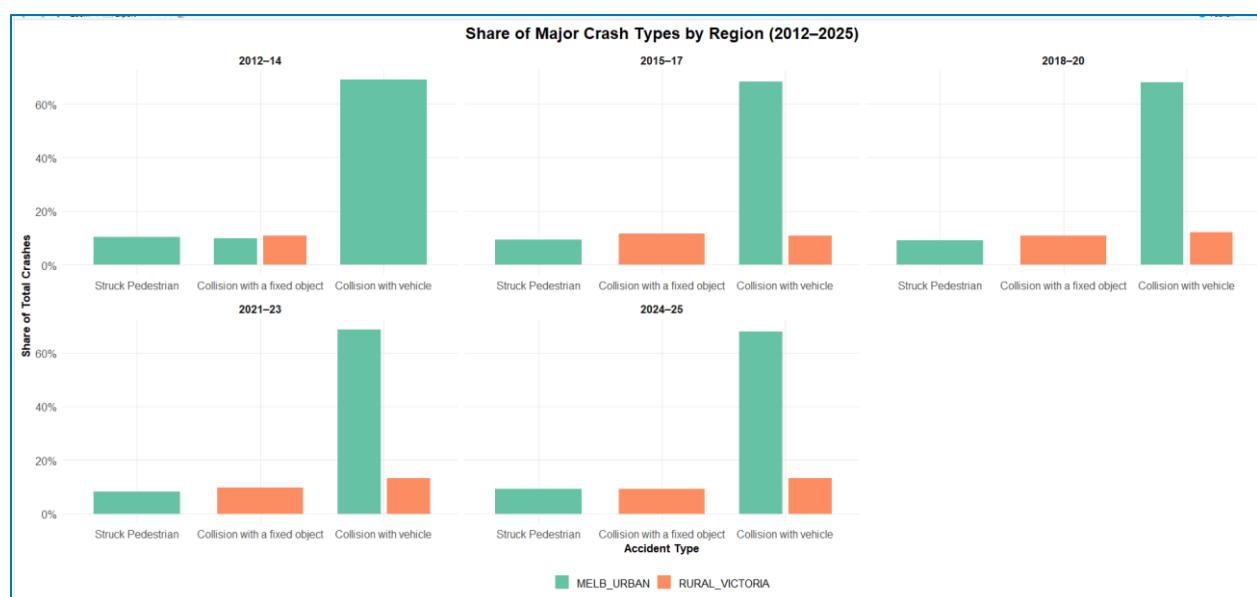


Figure 2.2.2.1 Share of Major Crash Types by region (2012-2025)

Crash type and regional patterns

Multi-vehicle collisions account for about two-thirds of all crashes in Melbourne's urban network, regardless of the time period. On the other hand, single-vehicle fixed-object collisions are more common in rural areas due to higher travel speeds and inadequate roadside protection. The pattern's structural stability confirms that rural isolation determines severity while urban density determines frequency.

Environmental Effects

The goal of this analysis was to determine how environmental factors affect the severity of crashes. The incident. To add the ROAD_SURFACE_COND_DESC attribute to every crash record, the csv and road_surface_cond.csv tables were combined on ACCIDENT_NO.

For Dry, Wet, Muddy, Icy, Snowy, and Unknown surfaces, faceted by period, grouped bars (geom_col(position = "dodge")) were plotted. To clearly display the proportions of fatalities, the y-axis was scaled using percent_format().

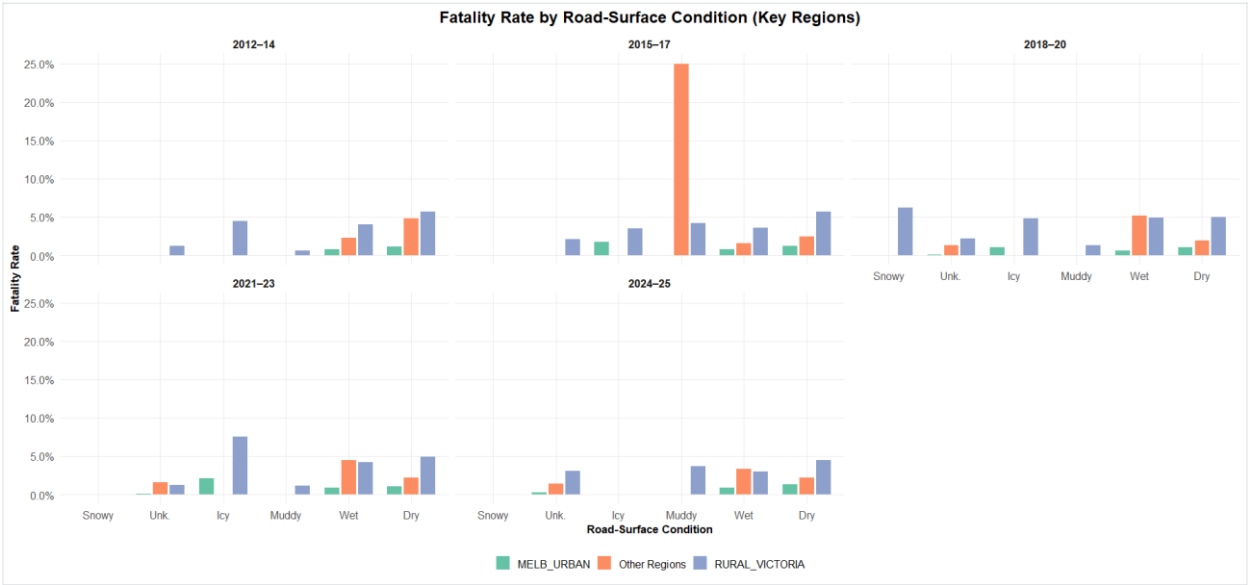


Figure 2.2.2.2 Environmental effects

The “Fatality Rate by Road-Surface Condition” visual revealed strong environmental sensitivity. Although most crashes occur on *dry* or *wet* roads, *icy* and *muddy* surfaces yield fatality rates up to 3–5× higher. The spike in 2015–2020 suggests a lag in regional surface maintenance or abnormal weather volatility. These conditions amplify driver-control loss, transforming otherwise survivable crashes into fatalities.

Cross-Factor Interactions

A two-way contingency summary was generated using group_by(ACCIDENT_TYPE_DESC, ROAD_SURFACE_COND_DESC, period) and summarise(fatal_share = sum (Fatal) / sum (Total)) to investigate the interaction effects between crash mechanisms and environmental conditions. Since a heatmap (geom_tile()) makes it easy to identify high-risk intersections, it was selected to visualise these co-occurrence patterns. High fatality share cells were indicated by dark blue

shading, which showed a strong correlation between icy/snowy surfaces and fixed-object crashes.

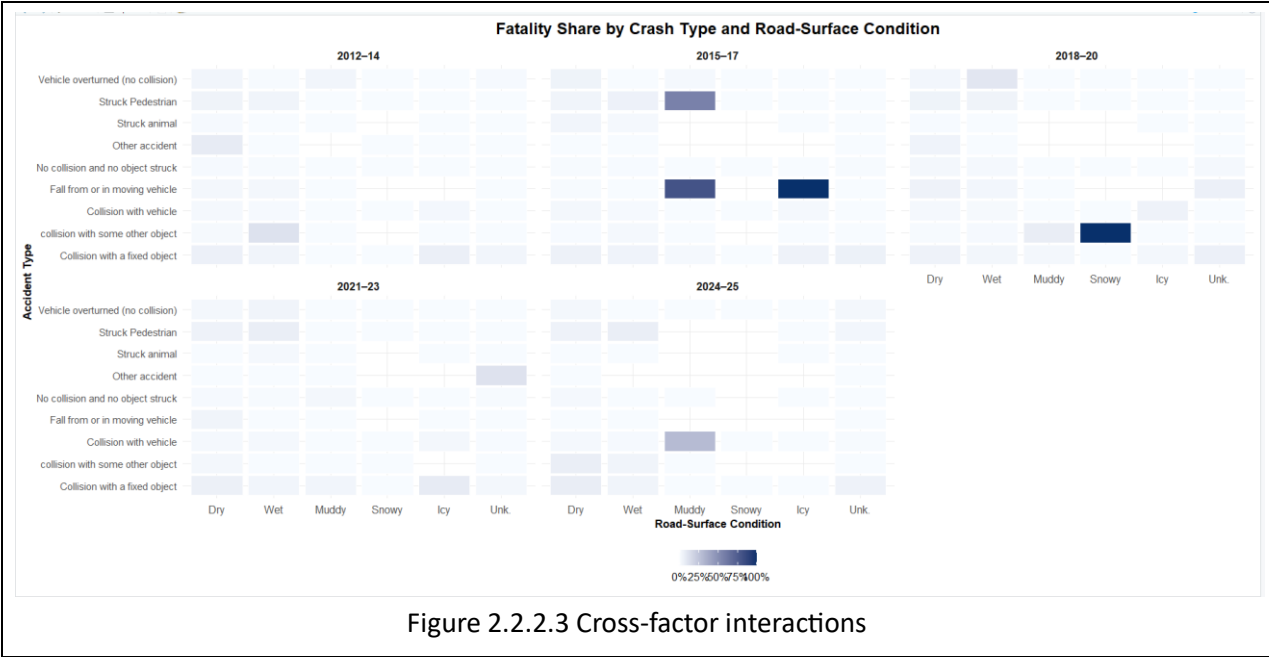
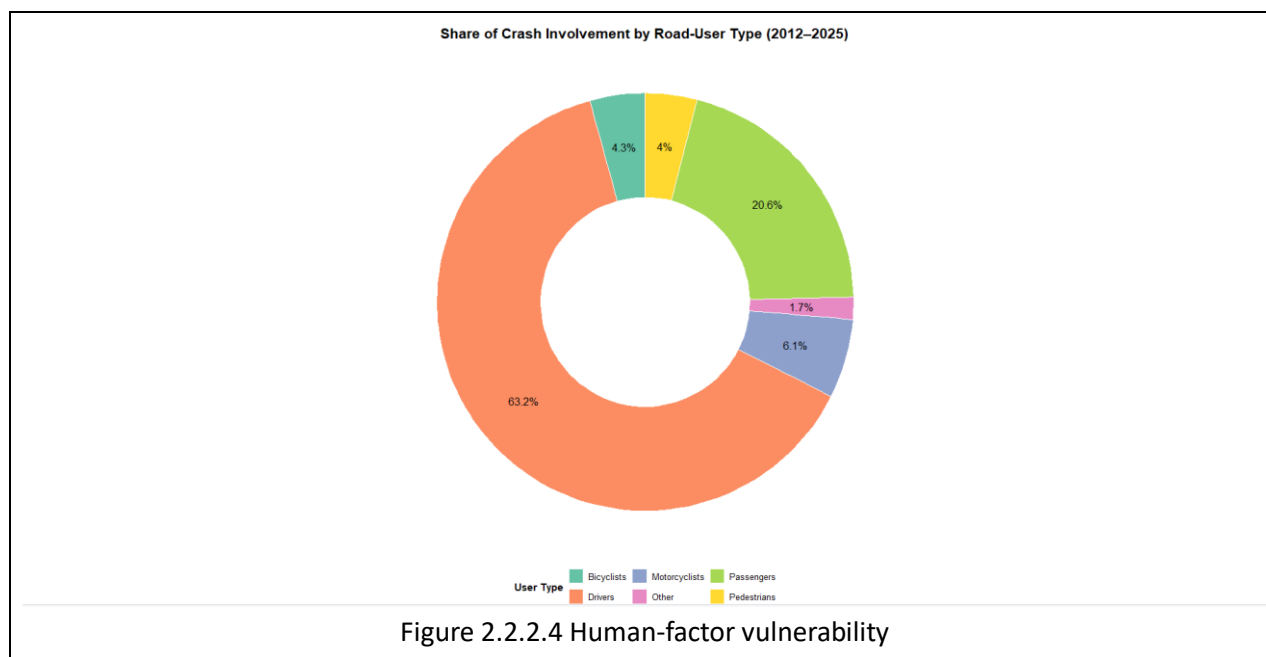


Figure 2.2.2.3 Cross-factor interactions

Vehicle-to-fixed-object crashes on muddy or icy surfaces, for instance, have the highest fatality shares when mechanical and environmental risks overlap, as shown by the heatmap visual (Crash Type × Surface Condition). These intersections attest to the fact that severity results from a combination of factors rather than from distinct causes.

Human-Factor Vulnerability

By differentiating between exposure and outcome severity, this visualisation measured vulnerability across user groups. Person-specific data was combined.CSV and mishap. ROAD_USER_TYPE_DESC and SEVERITY should be included in the csv on ACCIDENT_NO. To isolate death occurrences, only rows with SEVERITY == "Fatal" were kept. Pedestrians, motorcyclists, bicyclists, drivers, and passengers were the user types used to aggregate the results; other user types were grouped under "Other." Because it effectively communicates proportional comparisons and is appropriate for categorical fatality shares, a donut chart—a polar coordinate transformation of a bar plot—was chosen. The size of each segment indicated how much it contributed to the overall number of fatalities; position_stack (vjust = 0.5) was used to display textual percentage labels.



According to the donut chart, drivers account for the highest percentage of all fatalities, which is in line with their exposure levels. However, compared to their presence on the road, motorcyclists and pedestrians account for a disproportionately larger share of fatalities. This demonstrates that vulnerability, rather than just exposure, is a significant factor in determining the risk of death.

Overall, between 2015 and 2020, fatality rates increased dramatically, especially in rural Victoria, despite stable exposure and road-user composition. This implies that weather or transient infrastructure were the main causes rather than human behavior. After 2020, the Towards Zero initiative's implementation and advancements in rural emergency response correspond with the drop-in death rates.

2.2.3 Mitigation strategies

This integrated insight shows that mechanical patterns, environmental volatility, and human vulnerability are long-lasting structural factors that influence road safety outcomes in Victoria more than randomness. The fact that these relationships have remained stable for over ten years suggests that specific, context-specific interventions are necessary to achieve additional reductions in fatalities. The following are possible mitigation strategies to reduce accidents and fatalities:

Focus Area	Recommendation	Expected Impact
Urban corridors	AI-based congestion monitoring, distraction-reduction enforcement	Reduce high-volume collision frequency
Rural highways	Roadside barriers, lane widening, GPS-linked emergency response	Mitigate single-vehicle fatality severity
Weather-sensitive areas	Adaptive speed limits and real-time friction sensors	Minimise surface-related loss-of-control crashes
Vulnerable users	Pedestrian visibility zones, motorcycle safety awareness	Address high vulnerability per exposure

Table 2.2.3.1 Mitigation strategies

2.3 Why do Frankston, Kingston, and Colac Otway still record severe crashes despite state-wide safety gains, and which spatial, behavioural, age-related, or environmental factors explain this persistence?

In the second stage of the investigation, how geography and surrounding conditions influenced crash severity across Victoria from 2012 to 2025 was analysed. The analysis integrates statistical modelling, spatial visualisation, and hotspot assessment to clarify how road layout, weather, and driver behaviour collectively increase the likelihood of serious crashes. Each crash node is mapped using Leaflet, with colours showing severity (fatal, serious, or other) and marker size reflecting frequency. This visualisation highlights where collisions persist and intensify. A ranking of Victoria's ten most dangerous hotspots identifies intersections and corridors that consistently record severe outcomes.

Three sites emerge as the most critical: Frankston (Node 43940), Kingston (Node 264300), and Colac Otway (Node 44693). Together they represent the range of Victorian crash environments, from dense metropolitan corridors to remote rural routes. Frankston records the highest number of incidents, with 336 crashes (56 fatal and 140 serious) concentrated at the Frankston Dandenong Road and Cranbourne Road intersection. Kingston presents elevated risk along the Westall Road corridor, where inconsistent speeds and turning movements contribute to severity. Colac Otway, while less trafficked, demonstrates a higher fatality rate driven by isolation, speed, and challenging road geometry.

The comparative analysis that follows examines how each site's design, traffic exposure, and environmental conditions influence crash outcomes. The goal is to derive evidence-based strategies tailored to each setting: intersection upgrades for Frankston, improved traffic control for Kingston, and targeted safety enhancements for Colac Otway.

2.3.1 Methodology

2.3.1.1 Data Preparation

The data cleaning and preparation stage was carried out in RStudio. Each dataset `accident.csv`, `accident_location.csv`, `node.csv`, `vehicle.csv`, `road_surface_cond.csv`, `atmospheric_cond.csv`, `person.csv`, and `sub_dca.csv` was imported and checked for missing entries, inconsistent field names, and structural anomalies. A normalisation function (`norm()`) was developed to standardise column names across all tables, ensuring that key identifiers such as `ACCIDENT_NO` and `NODE_ID` were uniform for reliable joins. The function also created aligned variables where naming differed between datasets, for example converting `SEVERITY` to `accident_severity` and `ATMOSPH_COND` to `atmospheric_cond_desc`.

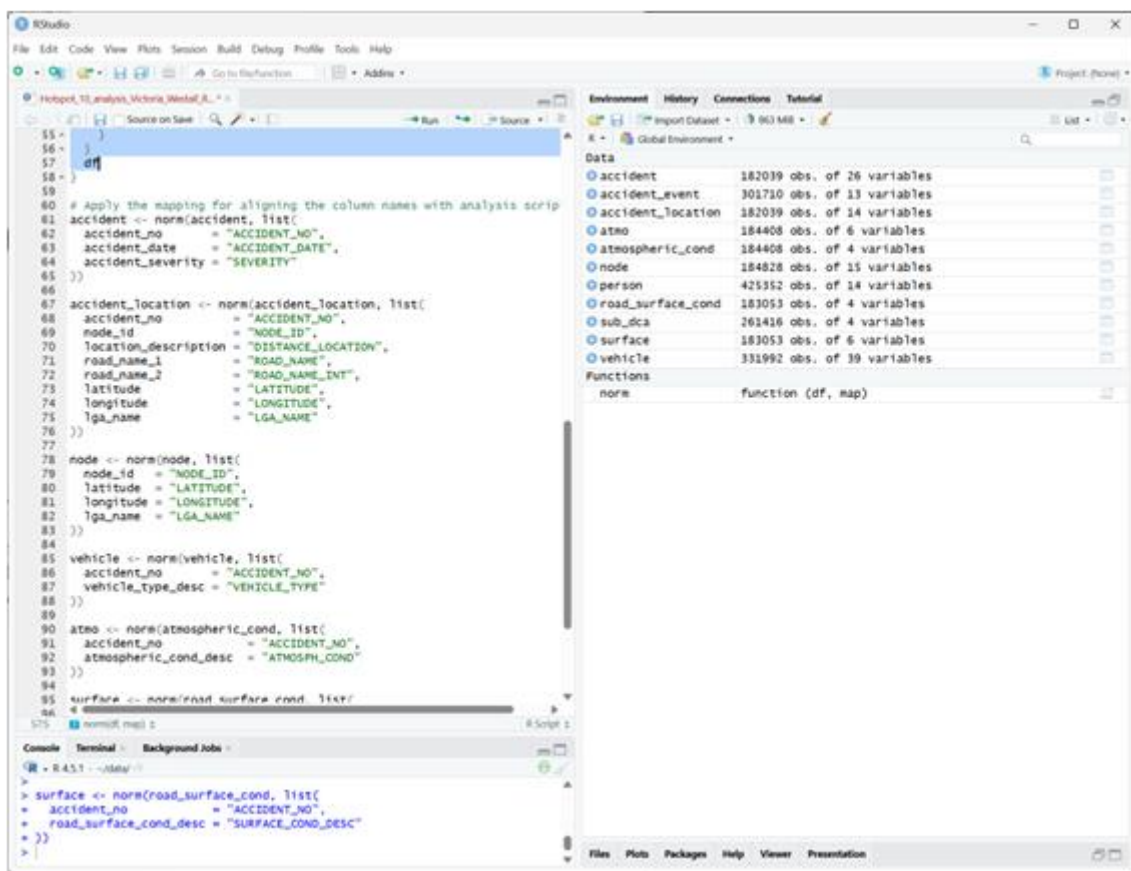


Figure 2.3.1.1.1 Data preparation in RStudio showing the normalisation process and verified datasets in the Environment pane.

2.3.1.2 Data Preparation

Following data cleaning, the individual tables were merged through the shared keys `ACCIDENT_NO` and `NODE_ID`, producing a single dataset at the crash level. Using this unified structure, an interactive map was built in Leaflet to visualise the spatial distribution of crashes across Victoria. Each crash node appeared as a colour coded marker: red for fatal, orange for serious, and yellow for minor incidents, with marker size scaled to the total number of recorded crashes at that point. The map serves as both orientation and evidence, allowing the geography of crash severity to be seen directly. By translating numerical data into spatial form, it reveals recurring corridors, hazardous intersections, and zones of persistent risk that raw tables conceal.

A companion ranking table highlights Victoria’s ten most fatal hotspots, providing the empirical foundation for detailed investigation. The highest risk location, Frankston (Node ID 43940), is examined next to illustrate how environmental and behavioural factors combine to drive severe outcomes.



Figure 2.3.1.2.1 Interactive Leaflet map of Victoria showing severe and fatal hotspots, highlighting Frankston Node 43940 (2012–2025)

node_id	lga_name	total_crashes	fatal_crashes	serious_crashes	severe_crashes	fatal_rate_per_1000	severe_rate_per_1000	weighted_risk
44693	COLAC OTWAY	56	16	24	40	285.7	714.3	1155.099747220331
1298	CARDINIA	49	14	28	42	285.7	857.1	1117.6649726508213
36276	BASS COAST	64	16	8	24	250	375	1043.5968174739091
264300	KINGSTON	64	16	16	32	250	500	1043.5968174739091
41977	KNOX	81	18	36	54	222.2	666.7	979.1730167
43940	FRANKSTON	336	56	140	196	166.7	583.3	970.2078244897386
49259	MILDURA	32	8	0	8	250	250	874.1268904
244459	WYNDHAM	32	8	0	8	250	250	874.1268904
50412	BASS COAST	70	14	0	14	200	200	852.5359754082631
40906	KNOX	144	24	36	60	166.7	416.7	829.6215148615097

Table 2.3.1.2.1 Top 10 fatal crash hotspots in Victoria (2012–2025)

2.3.2 Results

Crash Outcome Trends (2012–2025) for three highest hotspots

The temporal composition of crash severities across the three high fatality nodes, Frankston (43940), Colac Otway (44693), and Kingston (264300), reveals divergent safety trajectories. Frankston exhibits a sharp increase in the proportion of fatal crashes during 2015–2017, contrasting with a later reduction in serious injury share at Colac Otway and Kingston. Both

regional and metropolitan nodes show gradual transitions from fatal to non-fatal outcomes, suggesting the positive impact of targeted black spot treatments and road geometry improvements. Notably, the 2018–2020 and 2021–2023 periods coincide with the COVID-19 pandemic, when travel restrictions substantially reduced traffic exposure and total crash volumes, partially explaining the lower fatal shares and the clear dominance of serious or non-fatal incidents during these years.

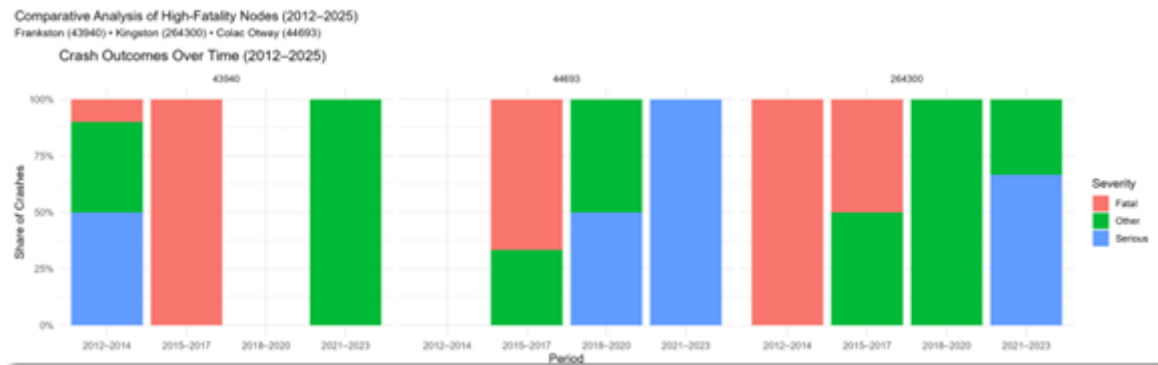


Figure 2.3.2.1 Crash severity composition over time for the three highest fatality nodes

These differences highlight the spatial and behavioural complexity of road risk. Frankston’s mid-period spike in fatalities likely reflects its higher traffic exposure and intersection density, while Colac Otway’s later improvement corresponds with rural infrastructure upgrades and reduced through-traffic volumes. Kingston’s relatively consistent trend suggests that metropolitan safety programs have effectively mitigated fatal outcomes, though serious injuries remain prominent. Collectively, these results illustrate that crash severity is shaped by the interplay of design, enforcement, and travel demand, reinforcing the need for location-specific interventions rather than uniform statewide measures.

Temporal Crash Patterns (2012–2025). Frankston Node 43940 and Comparative Hotspots.

Crash timing varies noticeably between Frankston (43940), Kingston (264300) and Colac Otway (44693). In Frankston, collisions tend to cluster between 2 p.m. and 6 p.m., aligning with the afternoon peak when roads are busier and visibility drops as the light fades. Kingston shows a two-part pattern, with crashes peaking in both the morning and late afternoon, mirroring commuter traffic flows typical of Melbourne’s suburban network. Colac Otway tells a different story: crashes there spread across the day but rise sharply after 10 p.m., a pattern consistent with driver fatigue, wildlife encounters and limited roadside lighting on rural routes.

A dip in crash counts during 2018–2020 reflects the COVID-19 lockdowns, when travel restrictions sharply reduced vehicle movement. The rebound seen in 2021–2023 suggests a return to normal traffic volumes. Taken together, the evidence shows that crash risk shifts with the rhythm of daily movement—urban congestion and glare drive afternoon risks, while rural fatigue and darkness shape night-time hazards. Effective responses should target these peaks through time-based patrols, better dusk lighting, and fatigue-awareness programs.

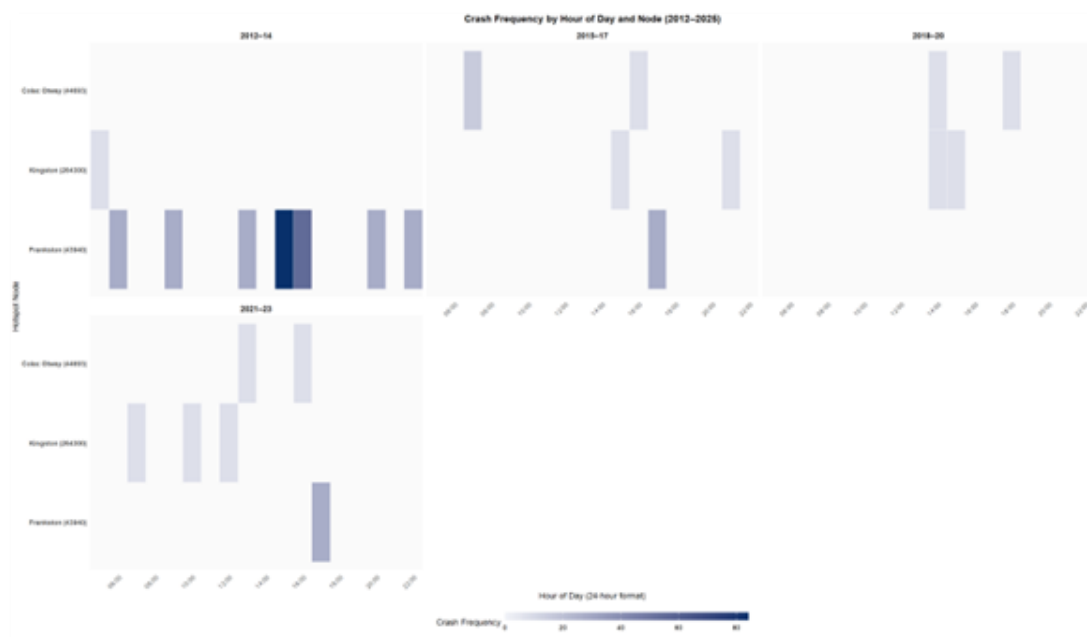


Figure 2.3.2.2 Crash frequency by hour of day and hotspot node (2012–2025), showing afternoon and evening peaks at Frankston, commuter-hour clusters at Kingston, and late-night risk at Colac Otway.

Age Distribution (2012–2025). For Frankston, Colac Otway, and Kingston node

The age composition of individuals involved in crashes across the three major hotspots reveals distinct demographic exposure patterns. Frankston (43940) shows a high proportion of crashes involving drivers aged 22–39, reflecting commuter and high-traffic profiles typical of metropolitan areas. Colac Otway (44693) presents a more balanced spread, though young adults (18–25) remain notably represented and consistent with rural risk profiles linked to higher speeds and limited enforcement presence. Kingston (264300) differs sharply, with a substantial share of older drivers (60+) involved, likely due to its suburban character and local trip patterns. Collectively, these results underline the need for age-specific safety interventions: youth-focused education and graduated-licensing reinforcement in regional zones, and mobility-support programs for ageing drivers in metropolitan corridors.

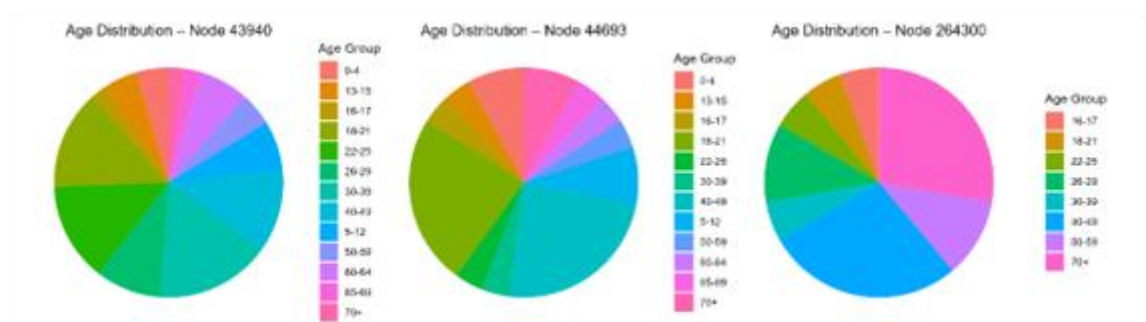


Figure 2.3.2.3 Age distribution of crash-involved persons at the three highest fatality nodes Frankston (43940), Colac Otway (44693), and Kingston (264300) showing differing demographic risk concentrations between young regional drivers and older metropolitan motorists.

2.3.3 Mitigation Strategies

Crash outcomes across Victoria show consistent spatial and temporal patterns shaped by road design, lighting, and driver behaviour. Frankston (43940) records afternoon and dusk peaks linked to congestion and reduced visibility, Kingston (264300) follows commuter-hour rhythms typical of suburban travel, and Colac Otway (44693) experiences late-night incidents tied to fatigue and limited lighting.

The fall in crashes during 2018–2020, followed by a rebound in 2021–2023, confirms that road risk rises predictably with traffic exposure once restrictions ease. These patterns underline that crash severity is largely structural, not random reinforcing the need for targeted, time-aware interventions such as dusk lighting upgrades, fatigue management, and commuter-period enforcement.

Focus Area	Observation	Insight / Interpretation	Implication for Mitigation
Temporal Patterns	Crash peaks between 14:00–18:00 at Frankston and Kingston; late-night clusters at Colac Otway.	Commuter congestion, reduced visibility, and driver fatigue drive these time-based risks.	Install dusk lighting, increase evening patrols, and expand fatigue-awareness programs.
Spatial Context	High crash density along arterial corridors and rural highways.	Persistent hotspots reveal infrastructure-related exposure rather than random distribution.	Prioritise black spot upgrades and realign high-risk intersections.
COVID-19 Impact	Crash volumes fell in 2018–2020 but rebounded post-2021.	Traffic reduction temporarily lowered exposure, confirming travel intensity as a key risk factor.	Implement adaptive traffic controls to manage congestion during peak hours.
Demographic Trends	Young regional drivers and older suburban motorists dominate severe outcomes.	Distinct behavioural and physiological vulnerabilities across age groups.	Targeted education for young drivers and mobility support for senior motorists.
Environmental Factors	Crashes intensify in wet or low-light conditions.	Visibility and traction loss contribute to loss-of-control incidents.	Expand adaptive lighting and use of real-time friction monitoring systems.

Table 2.3.3.1 Key Insights and Analysis (2012–2025)

3. Conclusion

According to the analysis, the higher crash fatality rates in rural Victoria are mostly structural rather than random. Most fatalities are caused by high-speed driving on mid-blocks and T-intersections along with poor surface quality; regular weather is only a minor factor. Although they are susceptible, motorcycles are not the primary cause of crashes; instead, heavy vehicles act as severity multipliers. The combined effects of road geometry, lighting, and driver behaviour over time and space are highlighted by persistent hotspots like Frankston, Kingston, and Colac Otway. These results highlight the need for location-specific interventions rather than one-size-fits-all regulations, such as surface maintenance, dusk or fatigue-focused enforcement, and speed management. To validate these findings and investigate how real-time data or telematics could further lower crash risk, future research could employ predictive modelling or machine-learning techniques.

4. Appendix

4.1 Abbreviations

ACCIDENT_NO: primary join key across accident-related tables.
DEG_URBAN_NAME: urbanisation/region label from node.csv.
MELB_URBAN: Melbourne Urban region bucket.
RURAL_VICTORIA: Rural Victoria region bucket.
SMALL_CITIES: Small Cities region bucket.
TOWNS: Towns region bucket.
ACCIDENT_TYPE_DESC: crash mechanism/type field.
ROAD_SURFACE_COND_DESC: road surface condition field.
ROAD_USER_TYPE_DESC: road-user category field.
SEVERITY: injury outcome indicator used for filtering fatalities.
CSV: comma-separated values files used for data sources.
km/h: kilometres per hour (unit used in speed bands).
R / ggplot2 / dplyr / leaflet: R language and packages used for analysis and visuals.
GPS: Global Positioning System (appears in mitigation).

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Figure 2.2.2.3: Cross-factor interactions.
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Table 2.1.3.1: Mitigation strategies.

Table 2.2.3.1: Mitigation strategies.

Table 2.3.3.1: Key insights and analysis (2012–2025).

5. References

1. VicRoads, "Crash Data Portal, 2012–2023," Department of Transport and Planning, Victoria, Australia, 2025.
2. H. Wickham, "dplyr: A Grammar of Data Manipulation," R Foundation for Statistical Computing, Vienna, Austria, 2024.